

HDOC Day-21 Activity

Menu-Driven C Program Using switch

Problem Statement

Write a menu-driven C program that accepts two positive integers, m and n, where $m \leq n$. Use a switch statement to perform the selected operation and continue displaying the menu until the user chooses Exit.

1. Print the first 10 multiples of each integer from m to n.
2. Print the addition table of each integer from m to n, using addends from 1 to 10.
3. Print the subtraction table of each integer from m to n, using subtrahends from 1 to 10.
4. Print the results obtained by dividing each integer from m to n by integers from 1 to 10, rounded to two decimal places.
5. Exit the program.

Algorithm

1. Start.
2. Declare integer variables m, n, choice, i and j.
3. Input two positive integers m and n.
4. Validate m as a positive integer; if invalid, display an error and ask again.
5. Input n and validate that n is positive and $m \leq n$; if invalid, display an error and ask again.
6. Display the menu with choices 1 to 5.
7. Input the user's choice and use a switch statement to perform the selected operation.
8. For choices 1 to 4, use nested loops: i runs from m to n and j runs from 1 to 10.
9. For division, use floating-point division and display the answer using %.2f.
10. If the menu choice is invalid, display an appropriate error message.
11. Repeat the menu until choice 5 is selected.
12. Stop.

Test-Case Table

Test	m	n	Choice	Expected Result	Type
------	---	---	--------	-----------------	------

1	2	3	1	Complete multiplication tables for 2 and 3 (1 to 10)	Valid
2	2	3	2	Complete addition tables for 2 and 3 (1 to 10)	Valid
3	2	3	3	Complete subtraction tables for 2 and 3 (1 to 10)	Valid
4	2	3	4	Complete division tables for 2 and 3 (1 to 10), to 2 decimal places	Valid
5	2	3	6	Invalid menu choice; menu displayed again	Invalid choice
6	0 then 2	3	5	Invalid positive-integer message; accepts corrected input	Invalid input
7	5	3 then 7	5	m > n error; accepts corrected n	Invalid range
8	5	5	1	Multiplication table of 5 (boundary m = n)	Boundary
9	2	3	5	Program ended.	Exit

C Program

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int m, n, choice, i, j;
```

```
    do
```

```
    {
```

```
        printf("Enter starting integer (m): ");
```

```
        scanf("%d", &m);
```

```
        if (m <= 0)
```

```
            printf("Invalid input. Enter a positive integer.\n");
```

```
    } while (m <= 0);
```

```
    do
```

```
    {
```

```
        printf("Enter ending integer (n): ");
```

```
        scanf("%d", &n);
```

```
        if (n <= 0)
```

```
            printf("Invalid input. Enter a positive integer.\n");
```

```
        else if (m > n)
```

```
            printf("Invalid input. m must be less than or equal to n.\n");
```

```
    } while (n <= 0 || m > n);
```

```
    do
```

```
    {
```

```
        printf("\n1. Multiplication Tables\n");
```

```
        printf("2. Addition Tables\n");
```

```
        printf("3. Subtraction Tables\n");
```

```
        printf("4. Division Tables\n");
```

```
        printf("5. Exit\n");
```

```
        printf("Enter your choice: ");
```

```
        scanf("%d", &choice);
```

```
        switch (choice)
```

```
        {
```

```
            case 1:
```

```
                for (i = m; i <= n; i++)
```

```

        {
            printf("\nMultiplication Table of %d\n", i);
            for (j = 1; j <= 10; j++)
                printf("%d x %d = %d\n", i, j, i * j);
        }
        break;

    case 2:
        for (i = m; i <= n; i++)
        {
            printf("\nAddition Table of %d\n", i);
            for (j = 1; j <= 10; j++)
                printf("%d + %d = %d\n", i, j, i + j);
        }
        break;

    case 3:
        for (i = m; i <= n; i++)
        {
            printf("\nSubtraction Table of %d\n", i);
            for (j = 1; j <= 10; j++)
                printf("%d - %d = %d\n", i, j, i - j);
        }
        break;

    case 4:
        for (i = m; i <= n; i++)
        {
            printf("\nDivision Table of %d\n", i);
            for (j = 1; j <= 10; j++)
                printf("%d / %d = %.2f\n", i, j, (float)i / j);
        }
        break;

    case 5:
        printf("Program ended.\n");
        break;

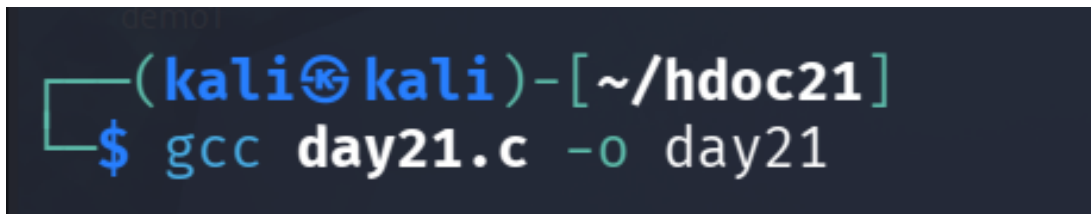
    default:
        printf("Invalid menu choice.\n");
}

} while (choice != 5);

return 0;
}

```

Compilation



Execution

1. Main Execution - All Menu Choices

Choices 1-2: Multiplication and Addition Tables

```
(kali㉿kali)-[~/hdoc21]
$ ./day21
Enter starting integer (m): 2
Enter ending integer (n): 3

1. Multiplication Tables
2. Addition Tables
3. Subtraction Tables
4. Division Tables
5. Exit
Enter your choice: 1

Multiplication Table of 2
2 x 1 = 2
2 x 2 = 4
2 x 3 = 6
2 x 4 = 8
2 x 5 = 10
2 x 6 = 12
2 x 7 = 14
2 x 8 = 16
2 x 9 = 18
2 x 10 = 20

Multiplication Table of 3
3 x 1 = 3
3 x 2 = 6
3 x 3 = 9
3 x 4 = 12
3 x 5 = 15
3 x 6 = 18
3 x 7 = 21
3 x 8 = 24
3 x 9 = 27
3 x 10 = 30

1. Multiplication Tables
2. Addition Tables
3. Subtraction Tables
4. Division Tables
5. Exit
Enter your choice: 2

Addition Table of 2
2 + 1 = 3
2 + 2 = 4
2 + 3 = 5
2 + 4 = 6
2 + 5 = 7
2 + 6 = 8
2 + 7 = 9
2 + 8 = 10
2 + 9 = 11
2 + 10 = 12

Addition Table of 3
3 + 1 = 4
3 + 2 = 5
3 + 3 = 6
3 + 4 = 7
3 + 5 = 8
3 + 6 = 9
3 + 7 = 10
3 + 8 = 11
3 + 9 = 12
3 + 10 = 13
```

Choices 3-5: Subtraction, Division and Exit

```
1. Multiplication Tables
2. Addition Tables
3. Subtraction Tables
4. Division Tables
5. Exit
Enter your choice: 3

Subtraction Table of 2
2 - 1 = 1
2 - 2 = 0
2 - 3 = -1
2 - 4 = -2
2 - 5 = -3
2 - 6 = -4
2 - 7 = -5
2 - 8 = -6
2 - 9 = -7
2 - 10 = -8

Subtraction Table of 3
3 - 1 = 2
3 - 2 = 1
3 - 3 = 0
3 - 4 = -1
3 - 5 = -2
3 - 6 = -3
3 - 7 = -4
3 - 8 = -5
3 - 9 = -6
3 - 10 = -7

1. Multiplication Tables
2. Addition Tables
3. Subtraction Tables
4. Division Tables
5. Exit
Enter your choice: 4

Division Table of 2
2 / 1 = 2.00
2 / 2 = 1.00
2 / 3 = 0.67
2 / 4 = 0.50
2 / 5 = 0.40
2 / 6 = 0.33
2 / 7 = 0.29
2 / 8 = 0.25
2 / 9 = 0.22
2 / 10 = 0.20

Division Table of 3
3 / 1 = 3.00
3 / 2 = 1.50
3 / 3 = 1.00
3 / 4 = 0.75
3 / 5 = 0.60
3 / 6 = 0.50
3 / 7 = 0.43
3 / 8 = 0.38
3 / 9 = 0.33
3 / 10 = 0.30

1. Multiplication Tables
2. Addition Tables
3. Subtraction Tables
4. Division Tables
5. Exit
Enter your choice: 5
Program ended.

(kali@kali)-[~/hdoc21]
$
```

2. Test-Case Executions - Additional Cases

Invalid Menu Choice

```
(kali㉿kali)-[~/hdoc21]
$ ./day21
Enter starting integer (m): 2
Enter ending integer (n): 3

1. Multiplication Tables
2. Addition Tables
3. Subtraction Tables
4. Division Tables
5. Exit
Enter your choice: 6
Invalid menu choice.

1. Multiplication Tables
2. Addition Tables
3. Subtraction Tables
4. Division Tables
5. Exit
Enter your choice: 5
Program ended.

(kali㉿kali)-[~/hdoc21]
$
```

Invalid Positive Integer Input

```
(kali㉿kali)-[~/hdoc21]
$ ./day21
Enter starting integer (m): 0
Invalid input. Enter a positive integer.
Enter starting integer (m): 2
Enter ending integer (n): 3

1. Multiplication Tables
2. Addition Tables
3. Subtraction Tables
4. Division Tables
5. Exit
Enter your choice: 5
Program ended.

(kali㉿kali)-[~/hdoc21]
$
```

Invalid Range ($m > n$)

```
(kali㉿kali)-[~/hdoc21]
$ ./day21
Enter starting integer (m): 5
Enter ending integer (n): 3
Invalid input. m must be less than or equal to n.
Enter ending integer (n): 7

1. Multiplication Tables
2. Addition Tables
3. Subtraction Tables
4. Division Tables
5. Exit
Enter your choice: 5
Program ended.

(kali㉿kali)-[~/hdoc21]
$ █
```

Boundary Case ($m = n$)

```
(kali㉿kali)-[~/hdoc21]
$ ./day21
Enter starting integer (m): 5
Enter ending integer (n): 5

1. Multiplication Tables
2. Addition Tables
3. Subtraction Tables
4. Division Tables
5. Exit
Enter your choice: 1

Multiplication Table of 5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50

1. Multiplication Tables
2. Addition Tables
3. Subtraction Tables
4. Division Tables
5. Exit
Enter your choice: 5
Program ended.

(kali㉿kali)-[~/hdoc21]
$ █
```

Result

The menu-driven C program was compiled and executed successfully using a switch statement and nested loops. It correctly prints multiplication, addition, subtraction and division tables for all integers from m to n , displays division results to two decimal places, validates positive input and the range $m \leq n$, handles invalid menu choices, and continues until the Exit option is selected.